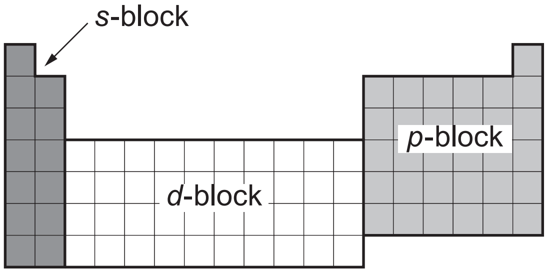
4. The Transition Metals



The elements in the Periodic Table can be divided into three blocks: the *s*-block, the *d*-block and the *p*-block.

The *s*-block and *d*-block contain only metallic elements. The *p*-block contains both metallic and non-metallic elements.

The transition metals are found within the *d*-block and are the metallic elements that serve as a bridge or 'transition' between the two sides of the table. They include the elements iron, silver, gold, vanadium and titanium.

The transition metals have similar properties to each other. These include high melting point, high density, good conductivity and malleability. The transition metals also have unique properties, making them different from the main group metals, including the ability to form compounds with different colours and different oxidation states.

Table 1 gives the properties of some transition metals and some Group 1 metals.

Element	Melting point (°C)	Boiling point (°C)	Oxidation states	Colours of compounds formed
iron	1536	2861	+2, +3	green and brown
lithium	180	1342	+1	white
potassium	63	760	+1	white
sodium	98	883	+1	white
titanium	1660	3287	+3, +4	violet and white
vanadium	1910	3407	+3, +4, +5	green, blue and yellow

Table 1

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(a) Tick (✓) the statement that **best** describes where metals are positioned within the Periodic Table. [1]

the *s*-block, *p*-block and *d*-block ☐

the *s*-block and *p*-block only ☐

the *s*-block only ☐

the *d*-block and *p*-block only ☐

the *p*-block only ☐

(b) Name the element in **Table 1** that is a liquid over the biggest temperature range. [1]

....

(c) Tick (✓) the **two** statements that correctly describe the oxidation states of the metals listed in **Table 1**. [2]

the Group 1 metals and transition metals all have a +1 oxidation state ☐

the transition metals all have a +3 oxidation state ☐

the Group 1 metals all have a +1 oxidation state ☐

iron and lithium have the same oxidation states ☐

the Group 1 metals and transition metals all have a +4 oxidation state ☐

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(d) Consider the following statement.

'It is possible to tell the Group 1 and transition metals apart by the colour of the compounds they form.'

Use the information in **Table 1** to give **one** reason to agree and **one** reason to disagree with this statement. [2]

Reason to agree ....

....

....

Reason to disagree ....

....

....

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